CSE 4137

Machine Learning
Module-02
Classification



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Topics: ML Classification

Week-05

- Introduction
- Logistic Regression
- Example

Logistic Regression: SML

Week-05, c-3

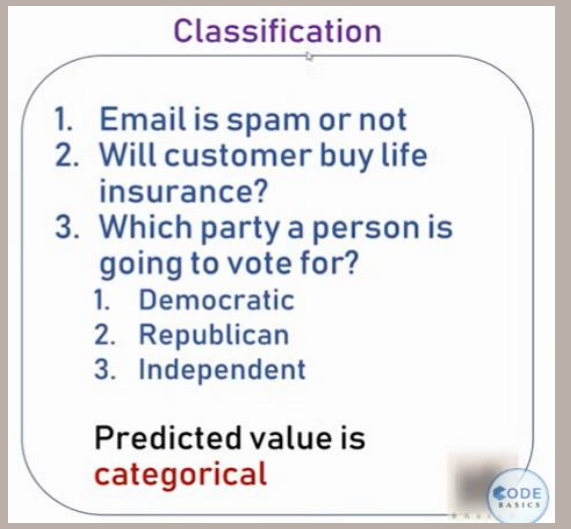
Multiple Linear Regression

•ogistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables.

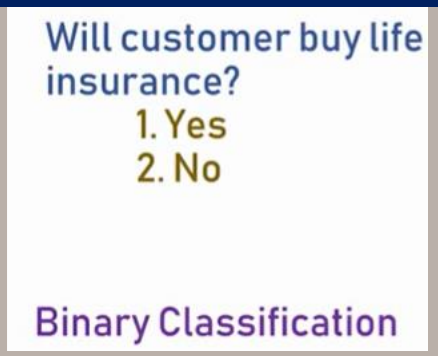
•Logistic regression predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, **it gives the probabilistic values which lie between 0 and 1**.

•Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas **Logistic regression is used for solving the classification problems**.





Types of Classification

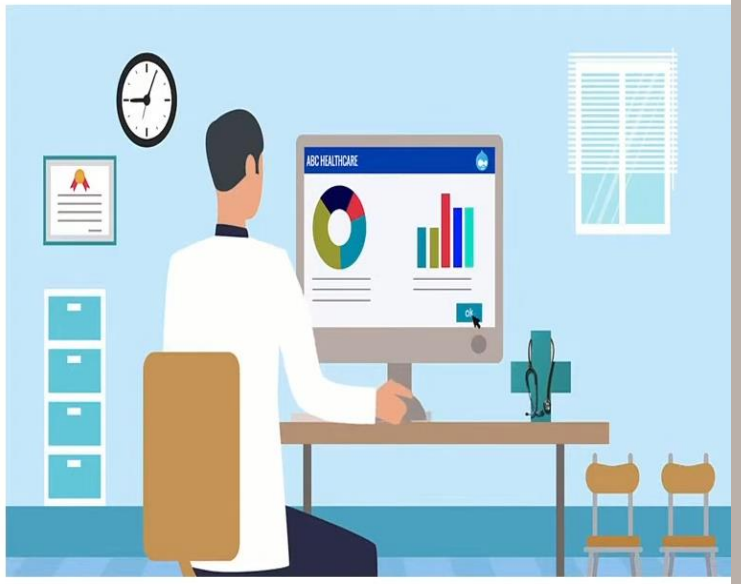




Logistic Regression: SML

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Real-life Example of Binary Classification



Heart Attack Prediction
Tumour Prediction

Predicts whether the patient has Heart Attack/ Tumour or not



Credit-Card Fraud

Check whether the transaction is fraud or not



Spam Detection

Check whether the email is spam or not

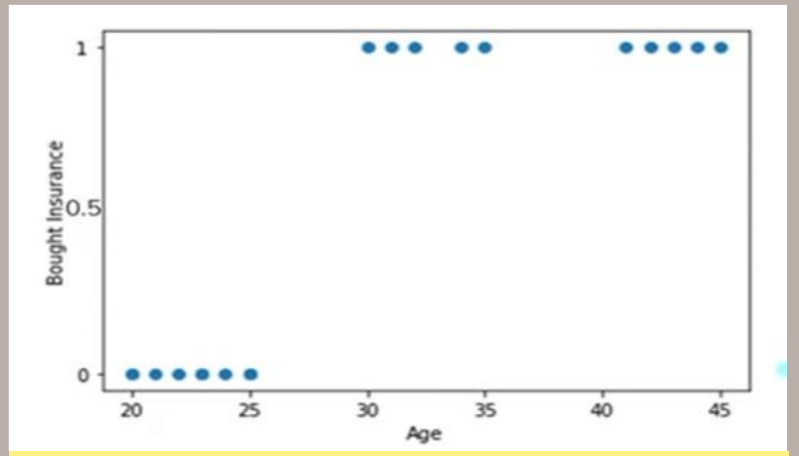
Logistic Regression: SML

Week-05, c-3

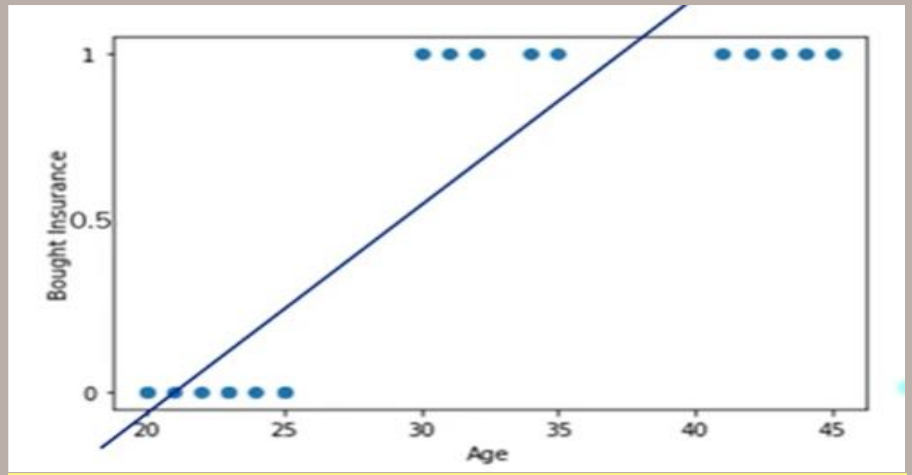
Example: Binary Classification

Prob: Predicts whether a person buy insurance or not with his age is given

Age	Bought Insurance
21	no
48	yes
32	yes
41	yes
20	no
35	yes
20	no
23	no



Scatter graph age vs Bought

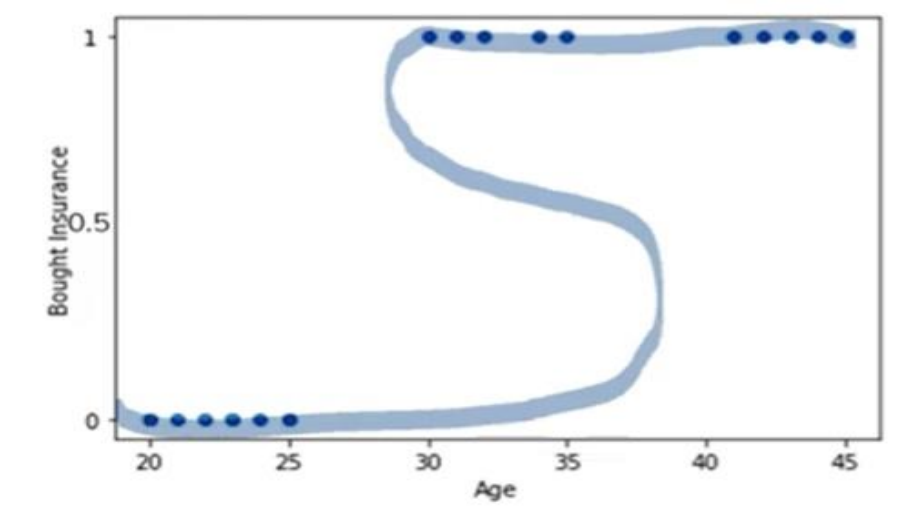


Linear Regression line not is fitting well

Logistic Regression: SML

Week-03, c-3

Logistic Regression graph

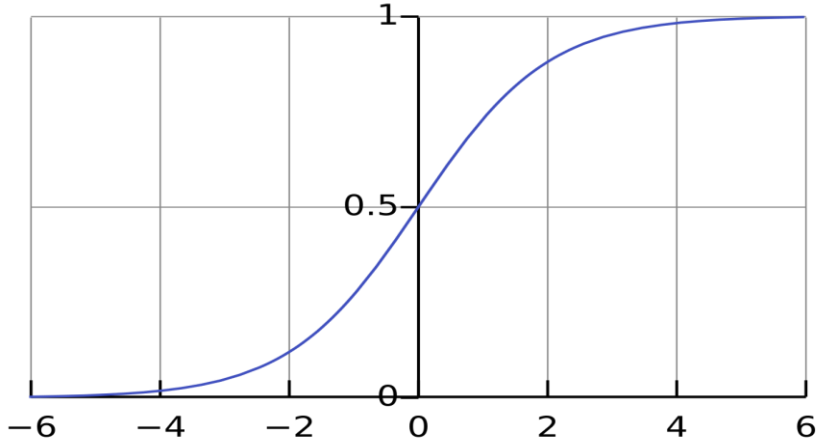


Logistic Regression graph fitting better than linear Regression graph.
This pattern can be obtain by Sigmoid Function

Bought should be in the range of 0 to 1
In the formula **Age→x** and **Bought→ σ(x)**

Sigmoid Function

A sigmoid function is any mathematical function whose graph has a characteristic S-shaped or sigmoid curve.



When $0<x<\alpha$ then $\sigma(x)\approx 1$ and when $\alpha<x<0$ then $\sigma(x)\approx 0$ [use the following formula]. So the value of x can be confined between 0 and 1 using Sigmoid function

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$Bought = \frac{1}{1+e^{-age}}$$

Logistic Regression: SML

Week-05, c-3

Confusion Matrix

A **confusion matrix** is a matrix that summarizes the performance of a machine learning model on a set of test data. It is a means of displaying the number of accurate and inaccurate instances based on the model's predictions. It is often used to measure the performance of classification models, which aim to predict a categorical label for each input instance.

The matrix displays the number of instances produced by the model on the test data.

- **True Positive (TP):** The model correctly predicted a positive outcome (the actual outcome was positive).
- **True Negative (TN):** The model correctly predicted a negative outcome (the actual outcome was negative).
- **False Positive (FP):** The model incorrectly predicted a positive outcome (the actual outcome was negative). Also known as a Type I error.
- **False Negative (FN):** The model incorrectly predicted a negative outcome (the actual outcome was positive). Also known as a Type II error.

Example: Confusion Matrix

- Consider the confusion matrix given below for a binary classifier predicting the presence of a disease
- The classifier made a total of 150 predictions Out of those 150 cases, the classifier predicted "yes" 100 times, and "no" 50 times.
- In reality, 100 patients in the sample have the disease, and 50 patients do not.

	Predicted No	Predicted Yes
Actual No	45	5
Actual Yes	5	95

Logistic Regression: SML

Week-05, c-3

Confusion Matrix

	Predicted No	Predicted Yes
Actual No	TN = 45	FP = 5
Actual Yes	FN = 5	TP = 95

- **Accuracy:** Overall, how often is the classifier correct?
- $Accuraccy = \frac{TN+TP}{TN+FP+FN+TP}$

$$= \frac{45+95}{150} = 93.33\%$$

- **Misclassification Rate:** Overall, how often is it wrong?
- $Missclassification\ Rate = \frac{FN+FP}{TN+FP+FN+TP}$

$$= \frac{5+5}{150} = 6.67\%$$

- **True Positive Rate:** When it's actually yes, how often does it predict yes?
- also known as "Sensitivity" or "Recall"
- $True\ Positive\ rate = \frac{TP}{Actual\ Yes}$

$$= \frac{95}{100} = 95\%$$

- **False Positive Rate:** When it's actually no, how often does it predict yes?
- $False\ Positive\ rate = \frac{FP}{Actual\ No}$

$$= \frac{5}{50} = 10\%$$

- **True Negative Rate:** When it's actually no, how often does it predict no?
- also known as "Specificity"
- $True\ Negative\ rate = \frac{TN}{Actual\ No}$

$$= \frac{45}{50} = 90\%$$

- **Precision:** When it predicts yes, how often is it correct?
- $Precision = \frac{TP}{Predicted\ Yes}$

$$= \frac{95}{100} = 95\%$$

- **Prevalence:** How often does the yes condition actually occur in our sample?
- $Prevalence = \frac{Actual\ Yes}{Total}$

$$= \frac{100}{150} = 66.67\%$$

Logistic Regression: SML

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Example: Confusion Matrix for Dog Image Recognition with Numbers

Index	1	2	3	4	5	6	7	8	9	10
Actual	Dog	Dog	Dog	Not Dog	Dog	Not Dog	Dog	Dog	Not Dog	Not Dog
Predicted	Dog	Not Dog	Dog	Not Dog	Dog	Dog	Dog	Dog	Not Dog	Not Dog
Result	TP	FN	TP	TN	TP	FP	TP	TP	TN	TN

- Actual Dog Counts = 6
- Actual Not Dog Counts = 4
- True Positive Counts = 5
- False Positive Counts = 1
- True Negative Counts = 3
- False Negative Counts = 1

		Predicted	
		Dog	**Not Dog**
Actual	**Dog**	True Positive (TP =5)	False Negative (FN =1)
	Not Dog	False Positive (FP=1)	True Negative (TN=3)

Logistic Regression: SML

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Example: Confusion Matrix for Dog Image Recognition with Numbers Implementaion with python

```
import numpy as np
from sklearn.metrics import
confusion_matrix,classification_report
import seaborn as sns
import matplotlib.pyplot as plt
```

```
actual = np.array(
  ['Dog','Dog','Dog','Not Dog','Dog','Not
Dog','Dog','Dog','Not Dog','Not Dog'])
predicted = np.array(
  ['Dog','Not Dog','Dog','Not
Dog','Dog','Dog','Dog','Dog','Not Dog','Not Dog'])
```

```
cm = confusion_matrix(actual,predicted)
cm
```

```
print(classification_report(actual, predicted))
```

```
[[5, 1],
 [1, 3]]
```

	precision	recall	f1-score	support
Dog	0.83	0.83	0.83	6
Not Dog	0.75	0.75	0.75	4
accuracy			0.80	10
macro avg	0.79	0.79	0.79	10
weighted avg	0.80	0.80	0.80	10

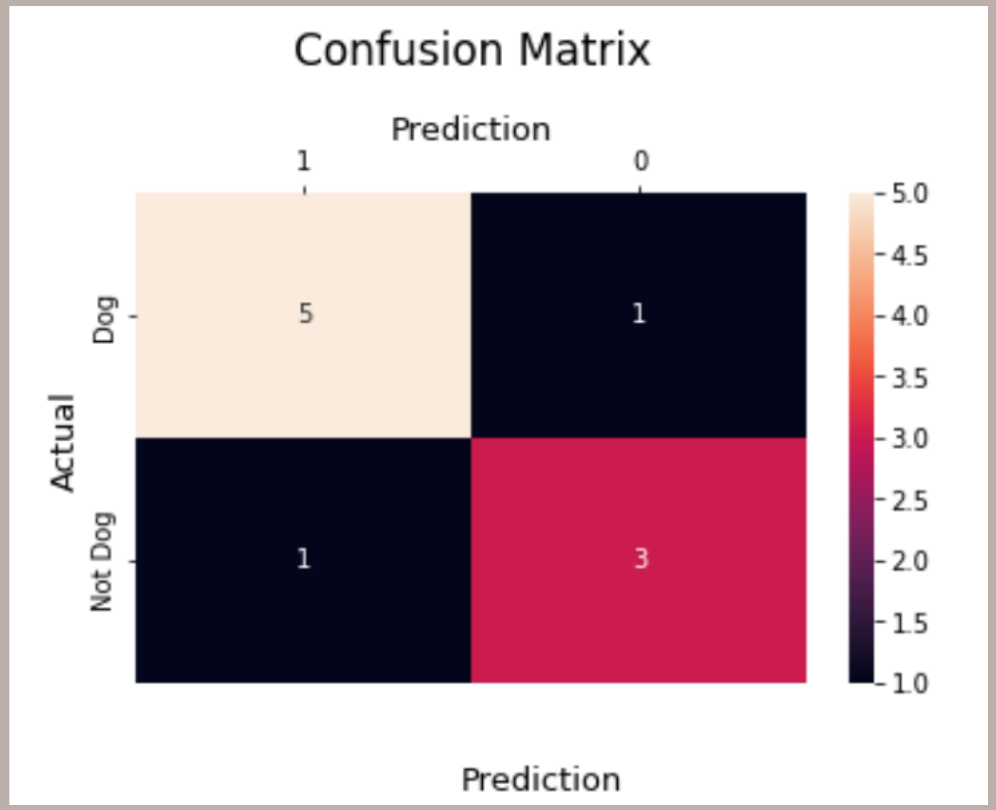
Logistic Regression: SML

Week-05, c-3

Example: Confusion Matrix for Dog Image Recognition with Numbers Implementaion with python

```
sns.heatmap(cm,
            annot=True,
            fmt='.0f',
            xticklabels=['1','0'],
            yticklabels=['Dog','Not Dog'])
plt.ylabel('Actual', fontsize=13)
plt.title('Confusion Matrix', fontsize=17, pad=20)
plt.gca().xaxis.set_label_position('top')
plt.xlabel('Prediction', fontsize=13)
plt.gca().xaxis.tick_top()

plt.gca().figure.subplots_adjust(bottom=0.2)
plt.gca().figure.text(0.5, 0.05, 'Prediction',
ha='center', fontsize=13)
plt.show()
```



Example: Logistic Regression

Logistic Regression: SML

Week-05, c-3

Example: Prediction of Insurance buying based on age

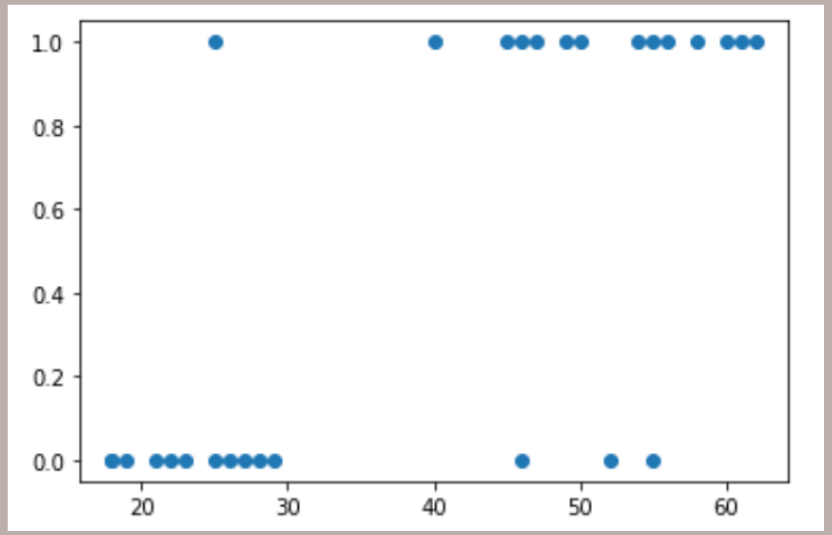
```
import numpy as np
import pandas as pd
from sklearn.metrics import
confusion_matrix,classification_report
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
```

```
df=pd.read_csv("F:\python\insurance_data.csv")
```

```
df['bought'].replace({'No':0,'Yes':1},inplace=True)
```

```
plt.scatter(x='age',y='bought',data=df)
```

```
#Separate Feature and Traget matrixs
X = df.iloc[:,:-1].values
y = df.iloc[:,1].values
```



Logistic Regression: SML

Week-05, c-3

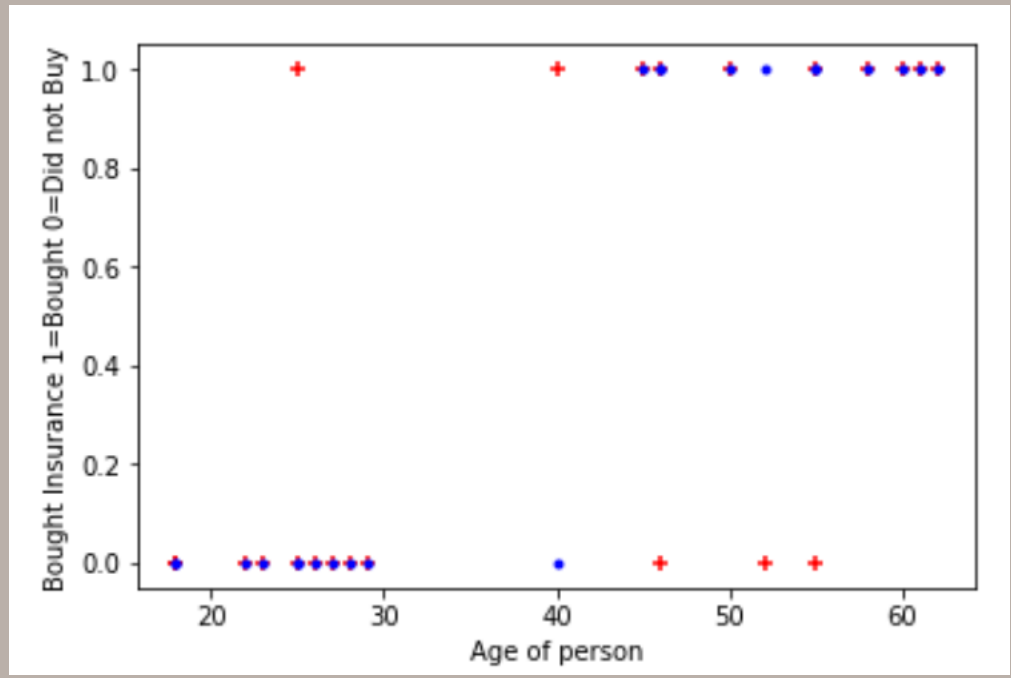
Example: Prediction of Insurance buying based on age

```
#Split the train and test dataset
X_train,X_test,y_train,y_test =
train_test_split(X,y,test_size = 0.2)
```

```
ml = LogisticRegression()
ml.fit(X_train,y_train)
```

```
y_pred = ml.predict(X_train)
# or use X_test inplace of X_train
```

```
plt.scatter(X_train,y_train,color= 'red', marker='+')
plt.scatter(X_train,y_pred,color='blue', marker='.')
plt.xlabel("Age of person")
plt.ylabel("Bought Insurance 1=Bought 0=Did not
Buy")
```



Logistic Regression: SML

Week-05, c-3

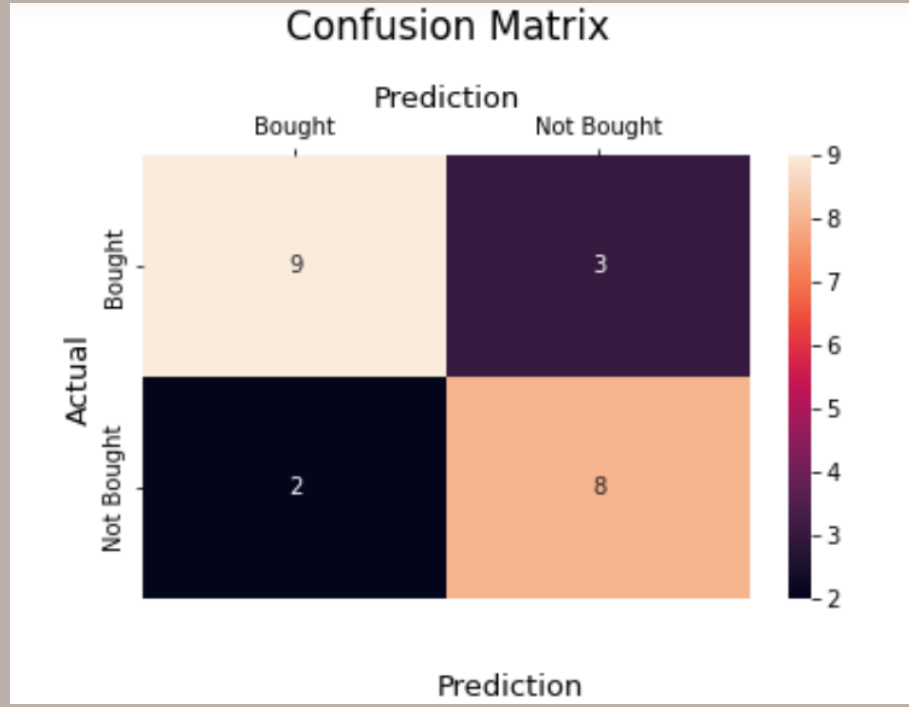
Example: Prediction of Insurance buying based on age

```
cm = confusion_matrix(y_train,y_pred)
```

```
sns.heatmap(cm,
            annot=True,
            fmt='.0f',
            xticklabels=['Bought','Not Bought'],
            yticklabels=['Bought','Not Bought'])
plt.ylabel('Actual', fontsize=13)
plt.title('Confusion Matrix', fontsize=17, pad=20)
plt.gca().xaxis.set_label_position('top')
plt.xlabel('Prediction', fontsize=13)
plt.gca().xaxis.tick_top()

plt.gca().figure.subplots_adjust(bottom=0.2)
plt.gca().figure.text(0.5, 0.05, 'Prediction',
ha='center', fontsize=13)
plt.show()
```

```
[[9, 3],
 [2, 8]]
```



End of Chapter